# sized_hdpe_500N

Hybrid Rocket Motor Design Report (HyProp)

Generated by HyProp

July 17, 2026

## 1 Mission Requirements

Required thrust: 500 N; total impulse: 4e+03 N s; mission apogee target: 3e+03 m; nozzle adaptation altitude: 1e+03 m (single fixed nozzle: optimum expansion is set at a fraction of the apogee, not at the apogee itself); safety factor: 1.5.

## 2 Inputs and Assumptions

Oxidizer: N2O at 293 K (self-pressurized). Fuel: HDPE. Injector: 4×1.65 mm, $C_d$=0.6 (SPI). Efficiencies: $\eta_{c^*}$=0.95, nozzle extra $\eta$=0.9673, divergence half-angle 15.0°.

## 3 Grain and Chamber Geometry

Geometry: cylindrical; length 508 mm; outer diameter 43.1 mm; port parameters: initial diameter 26.0 mm. Tank: 2.4 L, 1.66 kg oxidizer.

## 4 Transient Simulation

Burn time 17 s; total impulse 4.09e+03 N s; peak thrust 603 N; peak chamber pressure 38.1 bar; propellant used 2.05 kg; mean delivered $I_{sp}$ 203 s. Events: liquid runout at 7.26 s; pressure decay at 17.04 s.

## 5 Representative Operating Points

The plain time average is misleading for blowdown motors: it understates thrust by 51% and chamber pressure by 48% versus the impulse-weighted point. All design modules therefore use the weighted points below. Time share needed to deliver impulse fractions: 50% of $I_t$ in 21.3% of burn time, 75% of $I_t$ in 33.4% of burn time, 90% of $I_t$ in 41.3% of burn time, 95% of $I_t$ in 46.0% of burn time, 99% of $I_t$ in 56.5% of burn time.

### 5.1 Recommendations

- Nozzle design and reported performance: **impulse-weighted** point (each instant weighted by its contribution to total impulse).
- Thermal/erosion analysis: **heat-load** point (maximum of the Bartz $\sim P_c^{0.8}$ proxy).
- Structural sizing and materials: **peak-performance** envelope ($P_c$=38.1 bar, $T_c$=3176 K).
- Injector and feed system: **mass-flow-weighted** point.

Table 1: Representative operating points vs plain time average (key channels; full data in the XLSX export).

| channel | peak | imp-wtd | heat | mdot-wtd | median | time-avg |
|---|---|---|---|---|---|---|
| F [N] | 602.9 | 493.8 | 602.9 | 444.2 | 520.5 | 240 |
| Pc [bar] | 38.1 | 31.96 | 38.1 | 29.01 | 33.56 | 16.72 |
| O/F [-] | 6.443 | 5.359 | 6.443 | 4.915 | 5.535 | 3.003 |
| Isp [s] | 233.6 | 220.7 | 233.6 | 202.9 | 227.6 | 121.4 |
| c* [m/s] | 1520 | 1488 | 1520 | 1455 | 1510 | 1319 |
| Tc [K] | 3176 | 2963 | 3176 | 2807 | 3059 | 2171 |
| mdot_ox [kg/s] | 0.2278 | 0.1885 | 0.2278 | 0.1708 | 0.1975 | 0.09664 |
| mdot_f [kg/s] | 0.03535 | 0.03468 | 0.03535 | 0.03277 | 0.03568 | 0.02395 |

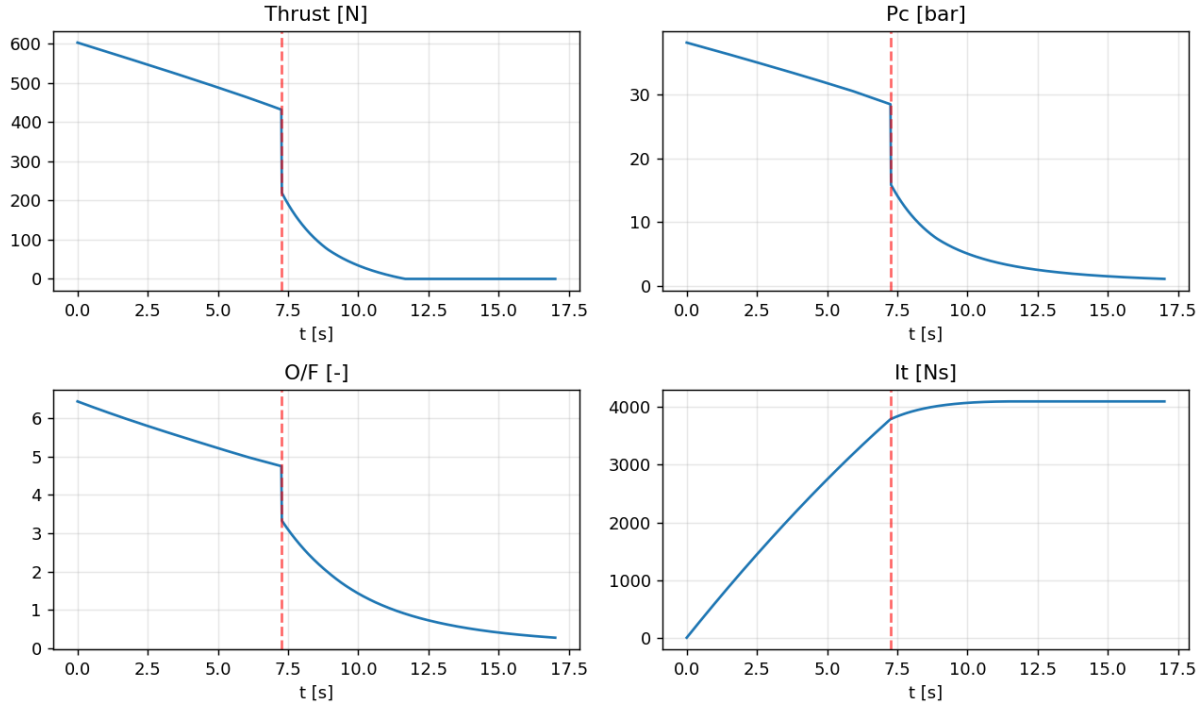
Table 2: Burn phases: liquid / transition / gas.

| phase | t0 [s] | t1 [s] | It [Ns] | It share | t share | F [N] | Pc [bar] | O/F |
|---|---|---|---|---|---|---|---|---|
| liquid | 0 | 7.24 | 3764 | 0.9216 | 0.4254 | 519.8 | 33.49 | 5.556 |
| transition | 7.26 | 7.48 | 47.75 | 0.01169 | 0.01293 | 225 | 16.18 | 3.351 |
| gas | 7.5 | 17.02 | 260 | 0.06366 | 0.5593 | 27.45 | 3.978 | 1.052 |

## 6 Nozzle and Performance

Throat diameter 11.7 mm; expansion ratio 5.02; impulse-weighted $I_{sp}$ 221 s, $c^*$ 1.49e+03 m/s, O/F 5.36.

## 7 Figures



## 8 Validation Note

Models validated against NASA-CEA-class thermochemistry (Cantera cross-check), HRAP transient benchmark (prusak_v2) and ESDU 91022/CoolProp N2O properties; see repository README. Empirical coefficients (a, n, $C_d$, $\eta$) require confirmation by fire tests before design freeze.